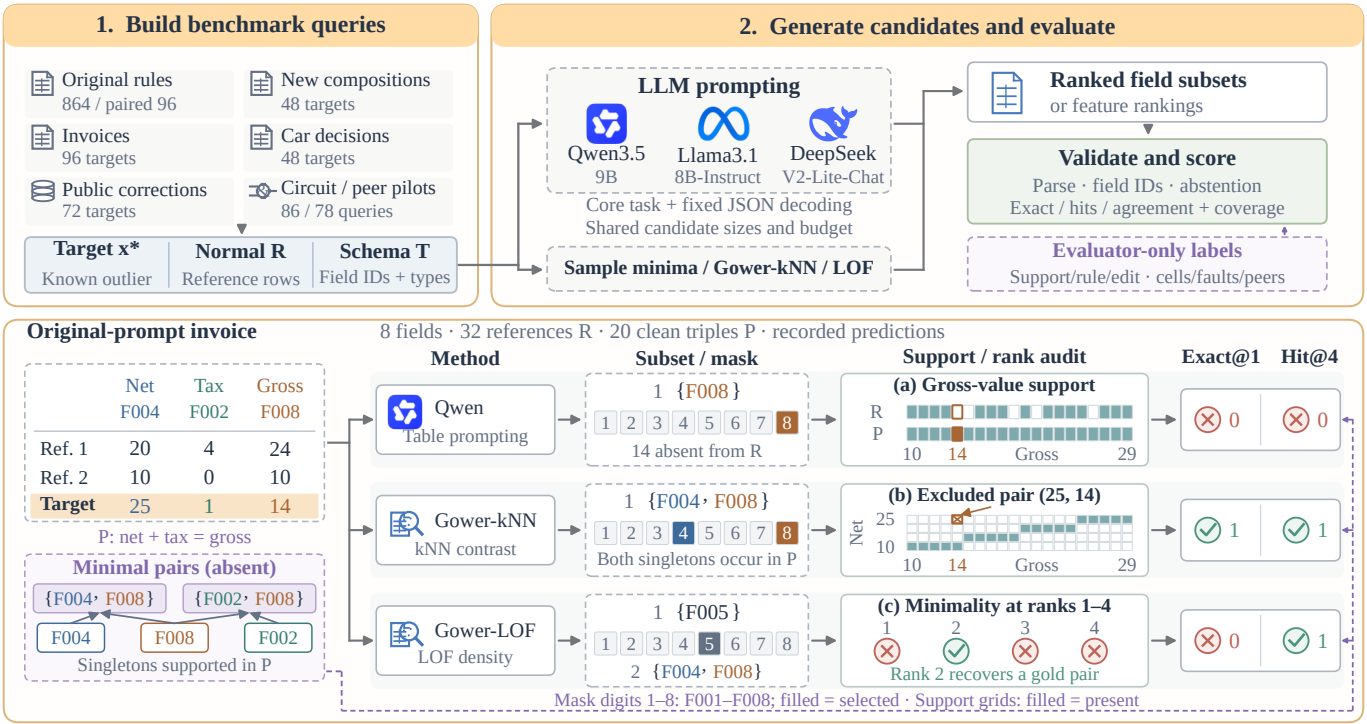

1. Build benchmark queries
Original rules
864 / paired 96
New compositions
48 targets
Invoices
96 targets
Car decisions
48 targets
Public corrections
72 targets
Circuit / peer pilots
86 / 78 queries
Target x*
Known outlier
Normal R
Reference rows
Schema T
Field IDs + types
2. Generate candidates and evaluate
LLM prompting
Qwen3.5
9B
Llama3.1
8B-Instruct
DeepSeek
V2-Lite-Chat
Core task + fixed JSON decoding
Shared candidate sizes and budget
Sample minima / Gower-kNN / LOF
Ranked field subsets
or feature rankings
Validate and score
Parse · field IDs · abstention
Exact / hits / agreement + coverage
Evaluator-only labels
Support/rule/edit · cells/faults/peers
Original-prompt invoice
8 fields · 32 references R · 20 clean triples P · recorded predictions
Net F004
Tax F002
Gross F008
Ref. 1 20 4 24
Ref. 2 10 0 10
Target 25 1 14
P: net + tax = gross
Minimal pairs (absent)
{F004, F008}
{F002, F008}
F004
F008
F002
Singletons supported in P
Method
Subset / mask
Support / rank audit
Exact@1
Hit@4
Qwen
Table prompting
1 {F008}
1 2 3 4 5 6 7 8
14 absent from R
(a) Gross-value support
R
P
10 14 Gross 29
0
0
Gower-kNN
kNN contrast
1 {F004, F008}
1 2 3 4 5 6 7 8
Both singletons occur in P
(b) Excluded pair (25, 14)
Net
25
10
10 14 Gross 29
1
1
Gower-LOF
LOF density
1 {F005}
1 2 3 4 5 6 7 8
2 {F004, F008}
(c) Minimality at ranks 1–4
1 2 3 4
Rank 2 recovers a gold pair
0
1
Mask digits 1–8: F001–F008; filled = selected · Support grids: filled = present